

Training Artificial Neural Networks (Part 1)

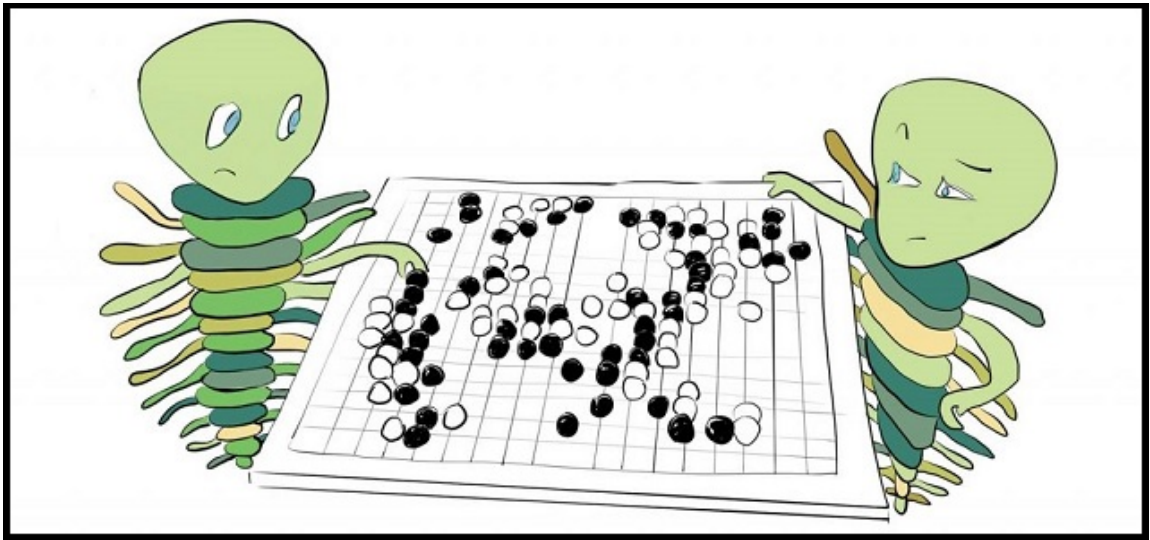
CSI 4106 - Fall 2026

Marcel Turcotte

Version: Aug 10, 2026 11:53

Preamble

Message of the Day



[From Turing's Chess to Neural Game Engines: AI in Video Games Today](#) by Alex Landa, ODSC, 2025-09-02, a [Podcast](#) (1h 8m) featuring Julian Togelius, an Associate Professor at New York University.

The text describes a podcast featuring Julian Togelius, an Associate Professor at New York University and co-director of the Game Innovation Lab. Togelius is the author of "Artificial Intelligence and Games," which has a second edition released in 2025. For those with access to a device on the University of Ottawa's network, a copy of this book is available for download.

- [Artificial Intelligence and Games](#)
 - [PDF](#), [EPUB](#)

I have been following Julian Togelius on social media for some time, although I have not yet had the opportunity to read his book.

Learning objectives

- **Explain** the architecture and function of feed-forward neural networks (FNNs).
- **Identify** common activation functions and understand their impact on network performance.
- **Introduce** a simple but functional implementation of a feed-forward neural networks.

Summary

3Blue1Brown (1/2)

<https://youtu.be/aircAruvnKk>

In my opinion, this is an **excellent** and **informative** video.

It is highly recommended that you watch this video. While it covers the concepts we have already explored, it presents the material in a manner that is challenging to replicate in a classroom setting.

- Provides a clear explanation of the **intuition** behind the effectiveness of neural networks, detailing the **hierarchy of concepts** briefly mentioned in the last lecture. (5m 31s to 8m 38s)
- Offers a compelling rationale for the necessity of a **bias term**.
- Similarly, elucidates the concept of **activation functions** and the importance of a squashing function.
- The segment beginning at 13m 26s offers a visual explanation of the **linear algebra** involved: $\sigma(WX^T + b)$.

3Blue1Brown (2/2)

<https://www.youtube.com/watch?v=IHZwWFHWa-w>

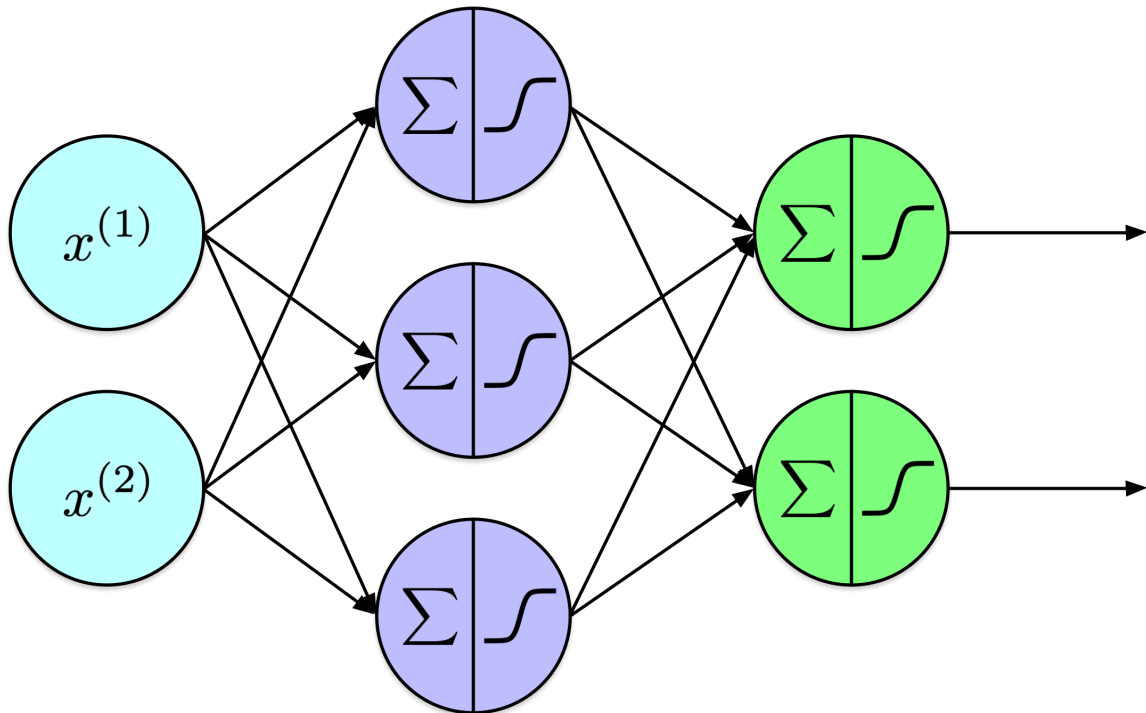
Our discussion will be on backpropagation. Viewing [Gradient descent, how neural networks learn](#) might be helpful.

We will revisit the concept of gradient descent in our discussion on the backpropagation algorithm. To review this topic, you can watch this video: [Gradient descent, how neural networks learn | Deep Learning Chapter 2](#) (duration: 20 minutes and 33 seconds).

Summary - DL

- **Deep learning (DL)** is a **machine learning** technique that can be applied to **supervised learning** (including **regression** and **classification**), **unsupervised learning**, and **reinforcement learning**.
- Inspired from the structure and function of **biological neural networks** found in animals.
- Comprises **interconnected neurons** (or units) arranged into **layers**.

Summary - FNN



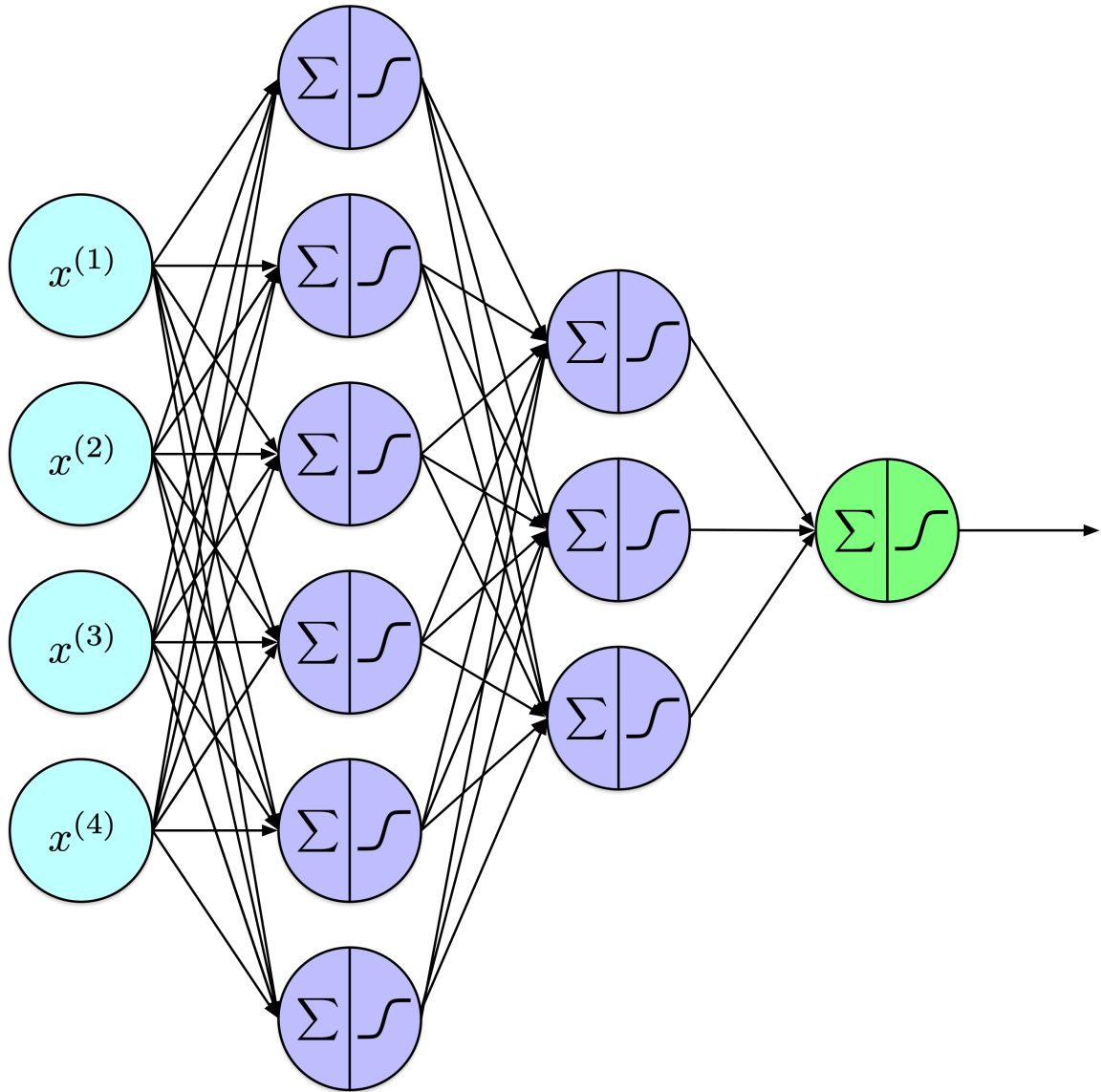
Information in this architecture flows unidirectionally—from left to right, moving from input to output. Consequently, it is termed a **feed-forward neural network (FNN)**.

Neural networks have **inputs** and **outputs**.

The network consists of **three layers**: input, hidden, and output. The **input layer** contains two nodes, the **hidden layer** comprises three nodes, and the **output layer** has two nodes. Additional hidden layers and nodes per layer can be added, which will be discussed later.

It is often useful to include explicit input nodes that do not perform calculations, known as **input units** or **input neurons**. These nodes act as placeholders to introduce input features into the network, passing data directly to the next layer without transformation. In the network diagram, these are the light blue nodes on the left. Typically, **the number of input units corresponds to the number of features**.

Summary - FNN



The number of **layers** and **nodes** can vary based on the specific requirements.

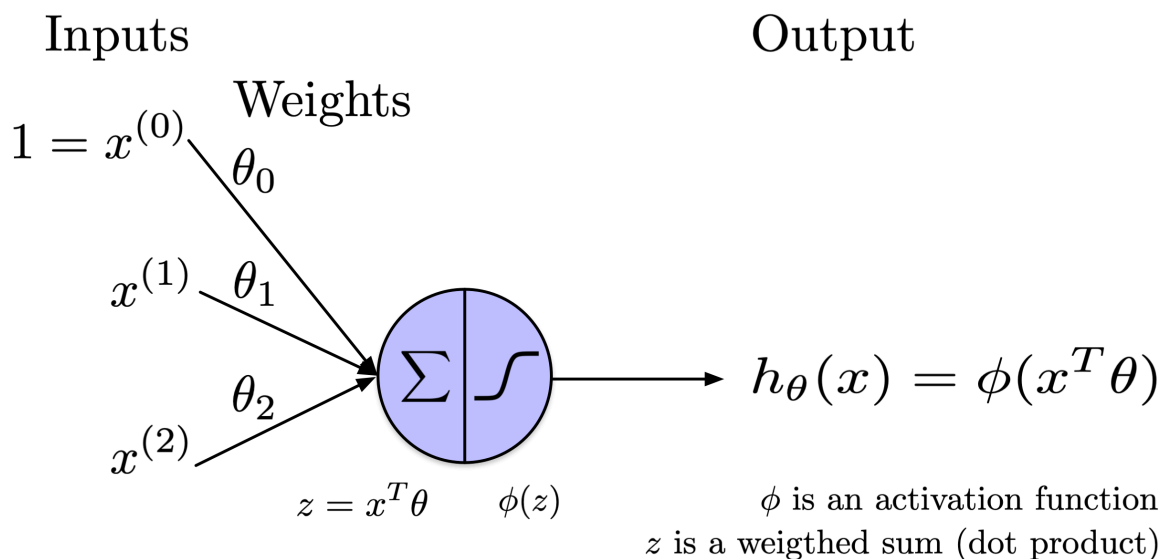
Neural networks can have a significantly large number of input nodes, often in the hundreds or thousands, depending on the complexity of the data. Additionally, they may contain numerous hidden layers. For instance, ResNet, which won the ILSVRC 2015 image classification task, features 152 layers. The authors of ResNet have demonstrated results for networks with 100 and even 1000 layers (He et al. 2016). However, the number of output nodes tends to be relatively small. In regression problems, there is typically one output node, while in classification tasks (whether multiclass or multilabel), the number of output nodes corresponds to the number of classes.

Consider a scenario in which one can determine the optimal number of layers and nodes for a neural network. Empirical evidence suggests that such networks excel in performing both classification and regression tasks. Despite the complexity arising from a large number of parameters, which complicates the interpretation of learned patterns,

understanding the forward pass, how the network generates predictions from new input data, is relatively straightforward.

Today's objective is to understand the process of adjusting the network's weights based on its current output. Specifically, we aim to understand how to utilize the output signal to propagate information backward through the network.

Summary - units



Introducing a fictitious input $x^{(0)} = 1$ is a *hack* that simplifies the expression $x^T \theta + b$.

In the diagram above, it is important to clarify that the inputs and output pertain specifically to this individual unit, rather than to the entire network's global inputs and output.

The name **activation** originates from the function's role in determining whether a neuron should be "activated" or "fired" based on its input.

Historically, the concept was inspired by biological neurons, where a neuron activates and transmits a signal to other neurons if its input exceeds a certain threshold. In artificial neural networks, the activation function serves a similar purpose by introducing non-linearity into the model. This non-linearity is crucial because it enables the network to learn complex patterns and representations in the data.

Common Activation Functions

```
In [2]: # Attribution: https://github.com/ageron/handson-ml3/blob/main/10\_neural\_net

import numpy as np
import matplotlib.pyplot as plt
```

```

from scipy.special import expit as sigmoid

def relu(z):
    return np.maximum(0, z)

def derivative(f, z, eps=0.000001):
    return (f(z + eps) - f(z - eps))/(2 * eps)

max_z = 4.5
z = np.linspace(-max_z, max_z, 200)

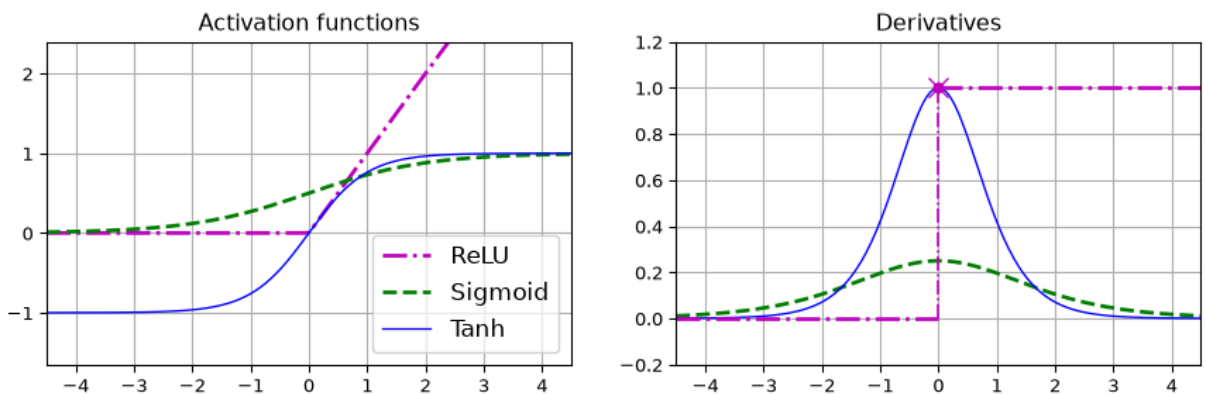
plt.figure(figsize=(11, 3.1))

plt.subplot(121)
plt.plot(z, relu(z), "m-.", linewidth=2, label="ReLU")
plt.plot(z, sigmoid(z), "g--", linewidth=2, label="Sigmoid")
plt.plot(z, np.tanh(z), "b-", linewidth=1, label="Tanh")
plt.grid(True)
plt.title("Activation functions")
plt.axis([-max_z, max_z, -1.65, 2.4])
plt.gca().set_yticks([-1, 0, 1, 2])
plt.legend(loc="lower right", fontsize=13)

plt.subplot(122)
plt.plot(z, derivative(sigmoid, z), "g--", linewidth=2, label="Sigmoid")
plt.plot(z, derivative(np.tanh, z), "b-", linewidth=1, label="Tanh")
plt.plot([-max_z, 0], [0, 0], "m-.", linewidth=2)
plt.plot([0, max_z], [1, 1], "m-.", linewidth=2)
plt.plot([0, 0], [0, 1], "m-.", linewidth=1.2)
plt.plot(0, 1, "mo", markersize=5)
plt.plot(0, 1, "mx", markersize=10)
plt.grid(True)
plt.title("Derivatives")
plt.axis([-max_z, max_z, -0.2, 1.2])

plt.show()

```



Géron (2022) – [10_neural_nets_with_keras.ipynb](#)

Consider the following observations:

- The sigmoid function produces outputs within the open interval $(0, 1)$.

- The hyperbolic tangent function ($\tanh$) has an image spanning the open interval $(-1, 1)$.
- The Rectified Linear Unit (ReLU) function outputs values in the interval $[0, \infty)$.

Additionally, note:

- The maximum derivative value of the sigmoid function is 0.25.
- The maximum derivative value of the $\tanh$ function is 1.
- The derivative of the ReLU function is 0 for negative inputs and 1 for positive inputs.

Furthermore:

- A node employing ReLU as its activation function generates outputs within the range $[0, \infty)$. However, its derivative, utilized in gradient descent during backpropagation, is constant, taking values of either 0 or 1.

Universal Approximation

The **universal approximation theorem** states that a feed-forward neural network with a single hidden layer containing a finite number of neurons can **approximate any continuous function** on a compact subset of $\mathbb{R}^n$, given appropriate weights and activation functions.

Cybenko (1989); Hornik et al. (1989)

The Universal Approximation Theorem (UAT) is a powerful theoretical assurance: in principle, a sufficiently wide single-hidden-layer network can approximate any continuous function. But it is not a practical prescription. In real problems, a deep architecture often achieves the same approximation accuracy with far fewer parameters and in a way that is more trainable and generalizable.

- Under relatively mild assumptions (e.g. non-polynomial activation, continuity, compact input domain), a feed-forward neural network with one hidden layer and a sufficiently large number of neurons (i.e. “wide enough”) can approximate any continuous function arbitrarily well (within arbitrarily small error) on a compact domain.
- The theorem is typically an *existence* result. It guarantees that such a network exists, but does not show how to find the right weights (i.e. the training procedure) or say how many neurons are needed precisely.
- The theorem also does not guarantee anything about generalization to unseen data (i.e. overfitting) or computational efficiency of training.
- The UAT says “there exists a wide enough network,” but it may require an extremely large number of neurons. In many practical settings, that becomes infeasible (too many parameters, too slow, risk of overfitting, etc.).

- Some functions are “hard” to approximate by shallow (i.e. single-hidden-layer) networks unless you use exponentially many neurons. In contrast, deeper networks may approximate the same function with far fewer parameters.
- UAT assumes you can pick the “right” weights. But in real training, optimization (e.g. via gradient descent) may get stuck in poor local minima, plateaus, saddle points, or fail to converge to the approximating solution.
- It gives no guarantee on how many training samples you need to realize a good approximation, or on generalization to new data.
- Even if a network can approximate a target function exactly (on training data), it may generalize poorly if the model is over-parameterized or if regularization is inadequate.
- UAT is silent on robustness to noise, stability, or extrapolation outside the training domain.

Naïve MLP

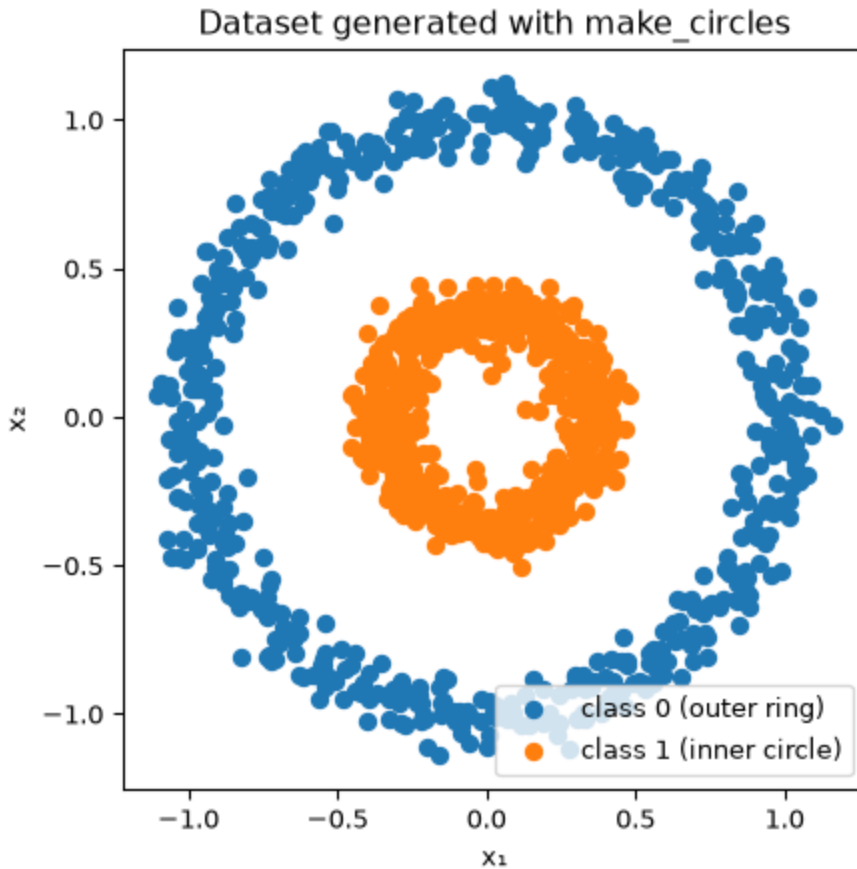
Data

```
In [3]: # Generate and plot the "circles" dataset
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles

# Generate synthetic data
X, y = make_circles(n_samples=1200, factor=0.35, noise=0.06, random_state=42)

# Separate coordinates for plotting
x1, x2 = X[:, 0], X[:, 1]

# Plot the two classes
plt.figure(figsize=(5, 5))
plt.scatter(x1[y==0], x2[y==0], color="C0", label="class 0 (outer ring)")
plt.scatter(x1[y==1], x2[y==1], color="C1", label="class 1 (inner circle)")
plt.xlabel("x1")
plt.ylabel("x2")
plt.title("Dataset generated with make_circles")
plt.axis("equal") # ensures circles look round
plt.legend()
plt.show()
```

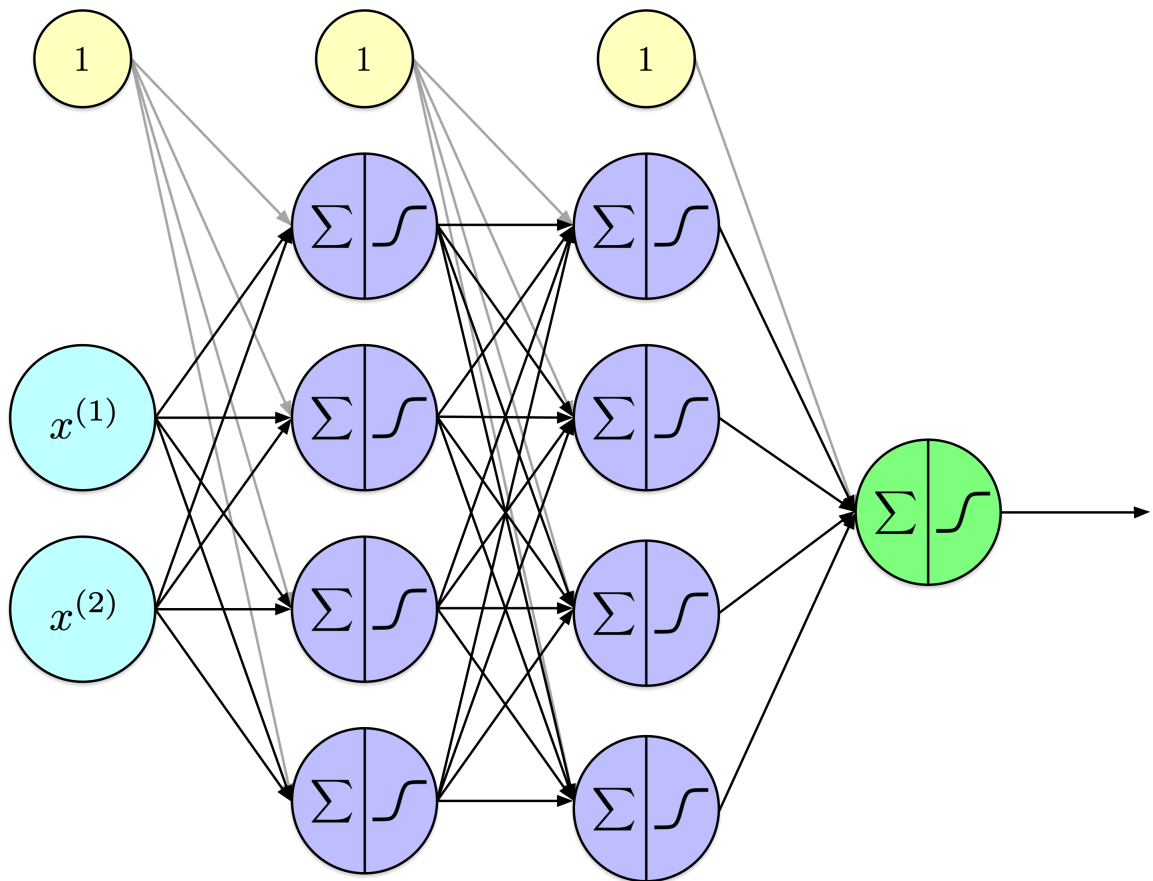



Concepts such as partial derivatives, gradient descent, and backpropagation can initially seem daunting. To mitigate this complexity, we propose an intermediary approach by constructing a simple yet fully operational neural network.

It is important to note that the proposed training algorithm is not intended to replace the standard back-propagation method used in practice.

We are constructing a dataset that comprises two distinct classes: an outer ring (class 1) and an inner circle (class 0). These classes are deliberately designed to be non-linearly separable within the (x_1, x_2) feature space, presenting a straightforward yet challenging scenario for classification tasks.

Architecture



Our neural network, a multi-layer perceptron, is designed with two input nodes corresponding to the two features present in our dataset. Through our experimentation using TensorFlow Playground and Keras, as detailed in [CircularSeparability](#), we determined that a configuration of two hidden layers containing four neurons each is effective for the classification of the samples.

The network produces a single output via the sigmoid activation function.

In the visualization, the black edges between units denote the model's weights. Additionally, we have included three nodes that output a constant value of 1 solely for visualization purposes; these nodes do not have a counterpart in the actual model. The gray edges connecting these nodes to other units represent the bias terms.

Utilities

```
In [4]: def sigmoid(z):
        return 1.0 / (1.0 + np.exp(-z))

def bce_loss(y_true, y_prob, eps=1e-9):

    """Binary cross-entropy loss (average over data)."""

    y_prob = np.clip(y_prob, eps, 1 - eps)
```

```
return -np.mean(y_true * np.log(y_prob) + (1 - y_true) * np.log(1 - y_pr
```

We are defining two utility functions, `sigmoid` and `bce_loss`. The binary cross-entropy loss has the same definition as that of our [logistic regression model](#). Do you remember another name for binary cross-entropy loss?

In `bce_loss`, the probabilities are constrained within the interval $[\epsilon, 1 - \epsilon]$. This clipping ensures that the logarithmic expressions $\log(0)$ and $\log(1 - 1) = \log(0)$ are avoided, thus preventing undefined or infinite values during computation.

Both functions are designed to take NumPy arrays (`ndarray`) as input. Namely, the `bce_loss` function calculates the loss for the entire dataset without requiring explicit iteration over individual data points.

NaïveMLP

The complete implementation is presented below and will be examined in the subsequent slides.

In [5]: `class NaiveMLP:`

```
"""
A minimal multilayer perceptron (MLP) utilizing a brute force training
algorithm that does not require derivative calculations.

Please note that the suggested training algorithm is intended solely for
didactic purposes and should not be mistaken for a genuine training algo
"""

def __init__(self, layer_sizes, step=0.1, seed=None):

    self.sizes = list(layer_sizes)
    self.step = float(step)
    rng = np.random.default_rng(seed)

    # Initialize weights and biases

    self.W = [rng.standard_normal(size=(in_d, out_d)) * 0.5
               for in_d, out_d in zip(layer_sizes[:-1], layer_sizes[1:])]

    self.b = [np.zeros(out_d) for out_d in layer_sizes[1:]]

def forward(self, X):

    """
    Simple forward pass: compute output activations only.

    X: shape (N, input_dim)

    Returns: output probabilities, shape (N,)
    """
```

```

"""

a = X
for W, b in zip(self.W, self.b):
    a = sigmoid(a @ W + b)

return a.ravel()

def predict(self, X, threshold=0.5):

    return (self.forward(X) >= threshold).astype(int)

def loss(self, X, y):

    return bce_loss(y, self.forward(X))

def _all_param_tags(self):

    """
    Yields tags referencing every scalar parameter:
    ('W', layer_idx, i, j) or ('b', layer_idx, j)
    """

    for l, W in enumerate(self.W):
        for i in range(W.shape[0]):
            for j in range(W.shape[1]):
                yield ('W', l, i, j)
            for j in range(self.b[l].shape[0]):
                yield ('b', l, j)

def _get_param(self, tag):
    kind = tag[0]
    if kind == 'W':
        _, l, i, j = tag
        return self.W[l][i, j]
    else:
        _, l, j = tag
        return self.b[l][j]

def _set_param(self, tag, val):
    kind = tag[0]
    if kind == 'W':
        _, l, i, j = tag
        self.W[l][i, j] = val
    else:
        _, l, j = tag
        self.b[l][j] = val

def train(self, X, y, epochs=10, verbose=True):

    """
    Simultaneous update:
    - For each scalar parameter  $\theta$ , try  $\theta + \delta$  for  $\delta$  in  $\{-step, 0, +step\}$ ,
      pick the  $\delta$  that gives minimal loss.
    - Collect all chosen  $\delta$ 's, then apply all updates together.
    """

```

```

for ep in range(1, epochs + 1):

    base_loss = self.loss(X, y)
    updates = {}

    # Probe all parameters
    for tag in self._all_param_tags():

        theta = self._get_param(tag)
        best_delta = 0.0
        best_loss = base_loss

        for delta in (-self.step, 0.0, +self.step):
            self._set_param(tag, theta + delta)
            trial_loss = self.loss(X, y)
            if trial_loss < best_loss:
                best_loss = trial_loss
                best_delta = delta

        # restore original
        self._set_param(tag, theta)
        updates[tag] = best_delta

    # Apply all deltas together
    for tag, d in updates.items():
        if d != 0.0:
            self._set_param(tag, self._get_param(tag) + d)

    new_loss = self.loss(X, y)

    if verbose:
        print(f"Epoch {ep:3d}: loss {base_loss:.5f} → {new_loss:.5f}")

    # optional early stop
    if abs(new_loss - base_loss) < 1e-12:
        break

```

To ensure the functionality of the resulting Jupyter Notebook, the whole class has been included here.

Class Definition

In [6]: **class** NaiveMLP:

"""

A minimal multilayer perceptron (MLP) utilizing a brute force training algorithm that does not require derivative calculations.

Please note that the suggested training algorithm is intended solely for didactic purposes and should not be mistaken for a genuine training algo

"""

Constructor

```
In [7]: def __init__(self, layer_sizes, step=0.1, seed=None):

        self.sizes = list(layer_sizes)
        self.step = float(step)
        rng = np.random.default_rng(seed)

        # Initialize weights and biases

        self.W = [rng.standard_normal(size=(in_d, out_d)) * 0.5
                   for in_d, out_d in zip(layer_sizes[:-1], layer_sizes[1:])]

        self.b = [np.zeros(out_d) for out_d in layer_sizes[1:]]
```

Memorizing the number of layers, the number of units per layer, and the learning algorithm's step size. Initializing the weights ($\bar{W}$) and biases (b).

Python

```
In [8]: seed = 0

rng = np.random.default_rng(seed)

layer_sizes = [2, 4, 4, 1]

[(in_d, out_d) for in_d, out_d in zip(layer_sizes[:-1], layer_sizes[1:])]

[(2, 4), (4, 4), (4, 1)]

...
```

```
In [9]: [rng.standard_normal(size=(in_d, out_d)) * 0.5 for in_d, out_d in zip(layer_

[array([[ 0.06286511, -0.06605243,  0.32021133,  0.05245006],
        [-0.26783469,  0.18079753,  0.65200002,  0.47354048]])],
array([[ -0.35186762, -0.63271074, -0.31163723,  0.02066299],
        [-1.16251539, -0.10939583, -0.62295547, -0.36613368],
        [-0.27212949, -0.15815008,  0.20581527,  0.52125668],
        [-0.06426733,  0.68323174, -0.33259734,  0.17575504]])],
array([[ 0.45173509],
        [ 0.04700615],
        [-0.37174962],
        [-0.46086269]])])]
```

How many weights has this network?

Python

```
In [10]: [out_d for out_d in layer_sizes[1:]]
```

```
[4, 4, 1]
```

```
...
```

```
In [11]: [np.zeros(out_d) for out_d in layer_sizes[1:]]
```

```
[array([0., 0., 0., 0.]), array([0., 0., 0., 0.]), array([0.])]
```

What is the number of bias terms in this network?

The network consists of a total of 37 parameters, which include 28 weights and 9 bias terms.

Forward Pass

```
In [12]: def forward(self, X):  
  
        """  
        Simple forward pass: compute output activations.  
        X: shape (N, input_dim)  
        Returns: output probabilities, shape (N,)  
        """  
  
        a = X  
        for W, b in zip(self.W, self.b):  
            a = sigmoid(a @ W + b)  
  
        return a.ravel()
```

Calculating the output activations. Determining the probability of each instance in the dataset X belonging to class 1.

The method proceeds sequentially layer-by-layer. In our running example, this involves three distinct processing layers.

What are W and b ?

W and b are lists, each containing three elements. The list W comprises weight matrices with dimensions 2×4 , 4×4 , and 4×1 , while b consists of bias arrays sized 4, 4, and 1, respectively.

What is the purpose of `zip(self.W, self.b)` ?

This function pairs each weight matrix with its corresponding bias array, resulting in three tuples, one for each of the second, third, and fourth layers.

What does X represent?

Leveraging `NumPy` makes the code compact, but it is important to recognize the underlying details. The parameter X encapsulates the entire dataset, comprising 200

samples with 2 features each. Within each iteration of the loop, the activations for all units in the current layer are computed for all examples.

Making predictions

```
In [13]: def predict(self, X, threshold=0.5):  
         return (self.forward(X) >= threshold).astype(int)
```

Indeed, predictions, plural.

The `forward` method produces an array of probabilities, each ranging from 0 to 1, indicating the likelihood that a particular example, x_i , belongs to class 1. By evaluating the expression `self.forward(X) >= threshold`, we obtain an array of boolean values: `True` if the probability exceeds the specified `threshold`, and `False` otherwise. These boolean values are subsequently converted into binary values of zeros and ones.

It is important to note that adjusting the decision boundary, `threshold`, allows for the manipulation of the precision-recall trade-off, providing flexibility in model performance.

Computing the loss

```
In [14]: def loss(self, X, y):  
         return bce_loss(y, self.forward(X))
```

The method in question calculates the loss for the entire dataset.

What would happen if we would return `bce_loss(y, self.predict(X))` instead of `bce_loss(y, self.forward(X))`?

Discussion

With the exception of the training algorithm, our neural network implementation is now complete.

For those who are not familiar with the back-propagation algorithm, how do you propose to learn the parameters of the model?

...

Change weights → compute loss → keep if better → repeat.

Pseudocode


```

pseudocode
for each epoch:
    deltas = {} # store best delta for each parameter
    for each parameter w in network:
        best_delta = 0
        best_loss = current_loss # loss computed with
current weights
        for delta in [-0.01, 0, +0.01]:
            w_temp = w + delta
            loss_temp = compute_loss_with_replacement(w_temp,
w, data)
            if loss_temp < best_loss:
                best_loss = loss_temp
                best_delta = delta
        deltas[w] = best_delta

    # Apply all updates simultaneously
    for each parameter w in network:
        w += deltas[w]

```

It is important to note that this algorithm is **fundamentally distinct from backpropagation**, which is widely used in practice. The primary reason for its introduction is its simplicity, notably the **absence of partial derivatives**.

The algorithm operates over several epochs, during which the predictive accuracy for the training set is incrementally enhanced with each iteration.

During each iteration, the algorithm evaluates whether to decrease, increase, or maintain the current value of each parameter.

Once the optimal adjustment for each parameter is determined, all changes are implemented simultaneously. This approach resembles the gradient descent technique discussed earlier in the semester and offers the advantage of straightforward parallelization.

Python

```

In [15]: def _all_param_tags(self):
        """
        Yields tags referencing every scalar parameter:
        ('W', layer_idx, i, j) or ('b', layer_idx, j)
        """
        for l, W in enumerate(self.W):
            for i in range(W.shape[0]):
                for j in range(W.shape[1]):
                    yield ('W', l, i, j)

```

```
for j in range(self.b[1].shape[0]):  
    yield ('b', l, j)
```

The above implements a generator, which is a Python concept that look simple, but packs a lot of power.

A **generator** is a kind of function that can *pause* its execution and *resume later*. It produces a sequence of values, one at a time, without storing them all in memory. You create one using the `yield` keyword.

Here is an example.

```
def countdown(n):  
    while n > 0:  
        yield n          # "yield" a value and pause  
        n -= 1
```

You can call it three times, then it will raise `StopIteration`.

```
c = countdown(3)  
print(next(c)) # 3  
print(next(c)) # 2  
print(next(c)) # 1  
try:  
    print(next(c))  
except StopIteration:  
    print("Caught StopIteration.")  
  
3  
2  
1  
Caught StopIteration.
```

Generators are often in `for` loops.

```
for value in countdown(3):  
    print(value)  
  
3  
2  
1
```

The functions `enumerate` and `zip` both return iterators, which function similarly to generators by facilitating lazy evaluation. This approach generates items dynamically during iteration, thereby avoiding the need to store all items in memory simultaneously.

Python

```
In [19]: class Demo:
```

```

def __init__(self, layer_sizes):
    self.sizes = list(layer_sizes)
    rng = np.random.default_rng(0)
    self.W = [rng.standard_normal(size=(in_d, out_d)) * 0.5
               for in_d, out_d in zip(layer_sizes[:-1], layer_sizes[1:])]
    self.b = [np.zeros(out_d) for out_d in layer_sizes[1:]]

def _all_param_tags(self):
    """
    Yields tags referencing every scalar parameter:
    ('W', layer_idx, i, j) or ('b', layer_idx, j)
    """

    for l, W in enumerate(self.W):
        for i in range(W.shape[0]):
            for j in range(W.shape[1]):
                yield ('W', l, i, j)
        for j in range(self.b[l].shape[0]):
            yield ('b', l, j)

def show(self):

    for tag in self._all_param_tags():
        print(tag)

```

```

In [20]: d = Demo([2,4,4,1])
         d.show()

```

```

('W', 0, 0, 0)
('W', 0, 0, 1)
('W', 0, 0, 2)
('W', 0, 0, 3)
('W', 0, 1, 0)
('W', 0, 1, 1)
('W', 0, 1, 2)
('W', 0, 1, 3)
('b', 0, 0)
('b', 0, 1)
('b', 0, 2)
('b', 0, 3)
('W', 1, 0, 0)
('W', 1, 0, 1)
('W', 1, 0, 2)
('W', 1, 0, 3)
('W', 1, 1, 0)
('W', 1, 1, 1)
('W', 1, 1, 2)
('W', 1, 1, 3)
('W', 1, 2, 0)
('W', 1, 2, 1)
('W', 1, 2, 2)
('W', 1, 2, 3)
('W', 1, 3, 0)
('W', 1, 3, 1)
('W', 1, 3, 2)
('W', 1, 3, 3)
('b', 1, 0)
('b', 1, 1)
('b', 1, 2)
('b', 1, 3)
('W', 2, 0, 0)
('W', 2, 1, 0)
('W', 2, 2, 0)
('W', 2, 3, 0)
('b', 2, 0)

```

Python

In [21]:

```

def _get_param(self, tag):
    kind = tag[0]
    if kind == 'W':
        _, l, i, j = tag
        return self.W[l][i, j]
    else:
        _, l, j = tag
        return self.b[l][j]

```

Given a `tag`, either `('W', l, i, j)` or `('b', l, j)`, the method retrieves the weight or the bias term corresponding to the provided indices. `l` designates the **layer**; `i` and `j` are row and column indices of `W[l]`, or `j` is an index in `b[l]`.

Python

```
In [22]: def _set_param(self, tag, val):
          kind = tag[0]
          if kind == 'W':
              _, l, i, j = tag
              self.W[l][i, j] = val
          else:
              _, l, j = tag
              self.b[l][j] = val
```

Similar logic, but the method updates `self.W[l][i, j]` or `self.b[l][j] = val`, depending on the type of tag.

Training (learning)

```
In [23]: def train(self, X, y, epochs=10, verbose=True):

          for ep in range(1, epochs + 1):

              base_loss = self.loss(X, y)
              updates = {}

              # Probe all parameters
              for tag in self._all_param_tags():

                  theta = self._get_param(tag)
                  best_delta = 0.0
                  best_loss = base_loss

                  for delta in (-self.step, 0.0, +self.step):
                      self._set_param(tag, theta + delta)
                      trial_loss = self.loss(X, y)
                      if trial_loss < best_loss:
                          best_loss = trial_loss
                          best_delta = delta

                  # restore original
                  self._set_param(tag, theta)
                  updates[tag] = best_delta

              # Apply all deltas together
              for tag, d in updates.items():
                  if d != 0.0:
                      self._set_param(tag, self._get_param(tag) + d)

              new_loss = self.loss(X, y)

              if verbose:
                  print(f"Epoch {ep:3d}: loss {base_loss:.5f} → {new_loss:.5f}")

              # optional early stop
```

```
if abs(new_loss - base_loss) < 1e-12:  
    break
```

"Simultaneous" $\approx$ a Jacobi-style step: you pick per-parameter deltas against the same baseline, then apply them all at once. Interactions between parameters are ignored while you're choosing them.

Ouf!

Does it work?

```
In [24]: from sklearn.model_selection import train_test_split  
from sklearn.metrics import accuracy_score  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.25, random_state=42, stratify=y  
)  
  
model = NaiveMLP([2, 4, 4, 1], step=0.06, seed=0)  
  
print("Initial loss:", model.loss(X_train, y_train))  
  
model.train(X_train, y_train, epochs=100)  
  
print("Train acc:", accuracy_score(y_train, model.predict(X_train)))  
print("Test acc: ", accuracy_score(y_test, model.predict(X_test)))
```

Initial loss: 0.7001969055705487
Epoch 1: loss 0.70020 → 0.69315
Epoch 2: loss 0.69315 → 0.69547
Epoch 3: loss 0.69547 → 0.69387
Epoch 4: loss 0.69387 → 0.69542
Epoch 5: loss 0.69542 → 0.69384
Epoch 6: loss 0.69384 → 0.69537
Epoch 7: loss 0.69537 → 0.69382
Epoch 8: loss 0.69382 → 0.69531
Epoch 9: loss 0.69531 → 0.69379
Epoch 10: loss 0.69379 → 0.69525
Epoch 11: loss 0.69525 → 0.69361
Epoch 12: loss 0.69361 → 0.69517
Epoch 13: loss 0.69517 → 0.69331
Epoch 14: loss 0.69331 → 0.69503
Epoch 15: loss 0.69503 → 0.69298
Epoch 16: loss 0.69298 → 0.69468
Epoch 17: loss 0.69468 → 0.69257
Epoch 18: loss 0.69257 → 0.69364
Epoch 19: loss 0.69364 → 0.69200
Epoch 20: loss 0.69200 → 0.69173
Epoch 21: loss 0.69173 → 0.69118
Epoch 22: loss 0.69118 → 0.68965
Epoch 23: loss 0.68965 → 0.68970
Epoch 24: loss 0.68970 → 0.68716
Epoch 25: loss 0.68716 → 0.68672
Epoch 26: loss 0.68672 → 0.68417
Epoch 27: loss 0.68417 → 0.68170
Epoch 28: loss 0.68170 → 0.67855
Epoch 29: loss 0.67855 → 0.67406
Epoch 30: loss 0.67406 → 0.66867
Epoch 31: loss 0.66867 → 0.66203
Epoch 32: loss 0.66203 → 0.65492
Epoch 33: loss 0.65492 → 0.64672
Epoch 34: loss 0.64672 → 0.63852
Epoch 35: loss 0.63852 → 0.62948
Epoch 36: loss 0.62948 → 0.61977
Epoch 37: loss 0.61977 → 0.60918
Epoch 38: loss 0.60918 → 0.59844
Epoch 39: loss 0.59844 → 0.58893
Epoch 40: loss 0.58893 → 0.57598
Epoch 41: loss 0.57598 → 0.56310
Epoch 42: loss 0.56310 → 0.55035
Epoch 43: loss 0.55035 → 0.53809
Epoch 44: loss 0.53809 → 0.52214
Epoch 45: loss 0.52214 → 0.50660
Epoch 46: loss 0.50660 → 0.49073
Epoch 47: loss 0.49073 → 0.47591
Epoch 48: loss 0.47591 → 0.45758
Epoch 49: loss 0.45758 → 0.44074
Epoch 50: loss 0.44074 → 0.42251
Epoch 51: loss 0.42251 → 0.41069
Epoch 52: loss 0.41069 → 0.38858
Epoch 53: loss 0.38858 → 0.36998
Epoch 54: loss 0.36998 → 0.35227
Epoch 55: loss 0.35227 → 0.34356

```
Epoch 56: loss 0.34356 → 0.32577
Epoch 57: loss 0.32577 → 0.31462
Epoch 58: loss 0.31462 → 0.29240
Epoch 59: loss 0.29240 → 0.27704
Epoch 60: loss 0.27704 → 0.25851
Epoch 61: loss 0.25851 → 0.25409
Epoch 62: loss 0.25409 → 0.23884
Epoch 63: loss 0.23884 → 0.23260
Epoch 64: loss 0.23260 → 0.21815
Epoch 65: loss 0.21815 → 0.21238
Epoch 66: loss 0.21238 → 0.19964
Epoch 67: loss 0.19964 → 0.19350
Epoch 68: loss 0.19350 → 0.17787
Epoch 69: loss 0.17787 → 0.16963
Epoch 70: loss 0.16963 → 0.15270
Epoch 71: loss 0.15270 → 0.14804
Epoch 72: loss 0.14804 → 0.13478
Epoch 73: loss 0.13478 → 0.13357
Epoch 74: loss 0.13357 → 0.12381
Epoch 75: loss 0.12381 → 0.12041
Epoch 76: loss 0.12041 → 0.10789
Epoch 77: loss 0.10789 → 0.10512
Epoch 78: loss 0.10512 → 0.09204
Epoch 79: loss 0.09204 → 0.08493
Epoch 80: loss 0.08493 → 0.07447
Epoch 81: loss 0.07447 → 0.07243
Epoch 82: loss 0.07243 → 0.06423
Epoch 83: loss 0.06423 → 0.06479
Epoch 84: loss 0.06479 → 0.06053
Epoch 85: loss 0.06053 → 0.05940
Epoch 86: loss 0.05940 → 0.05501
Epoch 87: loss 0.05501 → 0.05465
Epoch 88: loss 0.05465 → 0.05370
Epoch 89: loss 0.05370 → 0.05036
Epoch 90: loss 0.05036 → 0.04376
Epoch 91: loss 0.04376 → 0.04596
Epoch 92: loss 0.04596 → 0.04283
Epoch 93: loss 0.04283 → 0.04213
Epoch 94: loss 0.04213 → 0.04031
Epoch 95: loss 0.04031 → 0.03879
Epoch 96: loss 0.03879 → 0.03629
Epoch 97: loss 0.03629 → 0.03574
Epoch 98: loss 0.03574 → 0.03499
Epoch 99: loss 0.03499 → 0.03293
Epoch 100: loss 0.03293 → 0.03075
Train acc: 1.0
Test acc: 1.0
```

Our training algorithm is brittle. Obtaining these results required experimenting with `step` and `seed` .

Vizualization

In [25]: *# Plot helper: decision boundary in the original (x1, x2) plane*

```
def plot_decision_boundary(model, X, y, title="Naïve MLP decision boundary")

    # grid over the input plane
    pad = 0.3
    x1_min, x1_max = X[:,0].min()-pad, X[:,0].max()+pad
    x2_min, x2_max = X[:,1].min()-pad, X[:,1].max()+pad

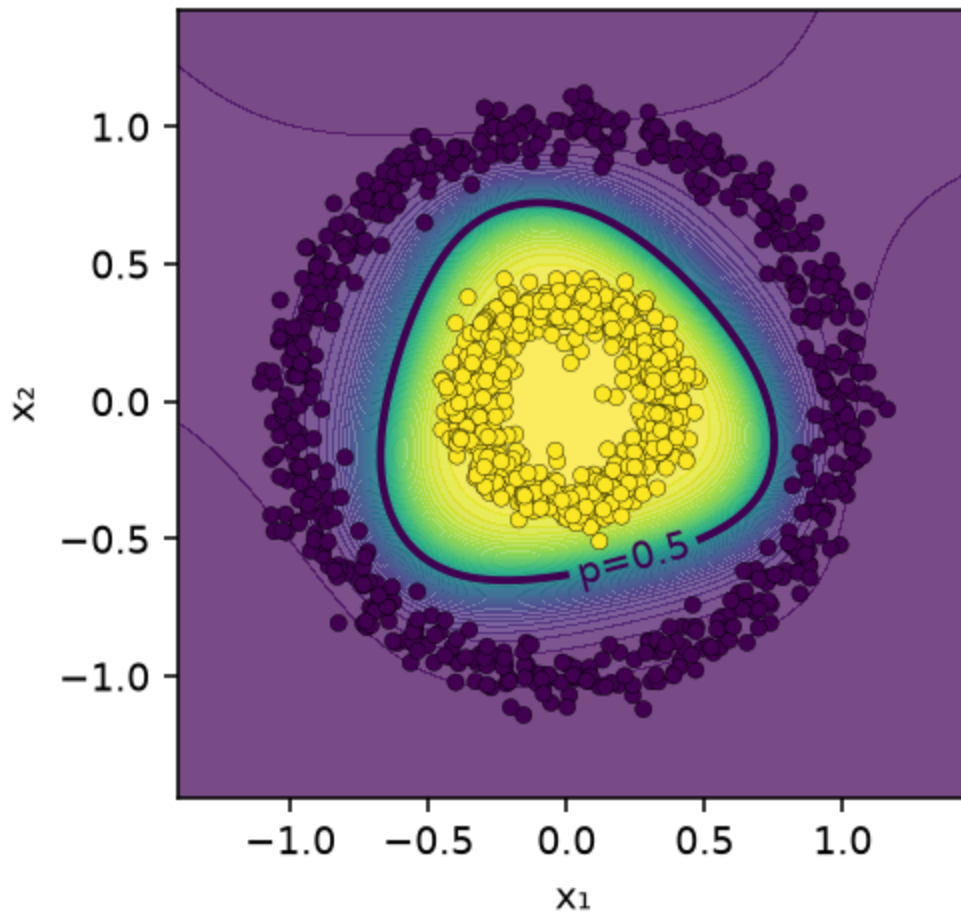
    xx, yy = np.meshgrid(
        np.linspace(x1_min, x1_max, 400),
        np.linspace(x2_min, x2_max, 400)
    )
    grid = np.c_[xx.ravel(), yy.ravel()]

    # predict probabilities on the grid
    p = model.forward(grid).reshape(xx.shape)

    # filled probabilities + p=0.5 contour + data points
    plt.figure(figsize=(3.75, 3.75), dpi=140)
    plt.contourf(xx, yy, p, levels=50, alpha=0.7)
    cs = plt.contour(xx, yy, p, levels=[0.5], linewidths=2)
    plt.scatter(X[:,0], X[:,1], c=y, s=18, edgecolor="k", linewidth=0.2)
    plt.clabel(cs, fmt={0.5: "p=0.5"})
    plt.title(title)
    plt.xlabel("x1")
    plt.ylabel("x2")
    plt.tight_layout()
    plt.show()

plot_decision_boundary(model, X, y)
```

Naïve MLP decision boundary



XOR-like data

```
In [26]: n_samples = 800
rng = np.random.default_rng(42)

X = rng.uniform(-6, 6, size=(n_samples, 2))
x1, x2 = X[:, 0], X[:, 1]

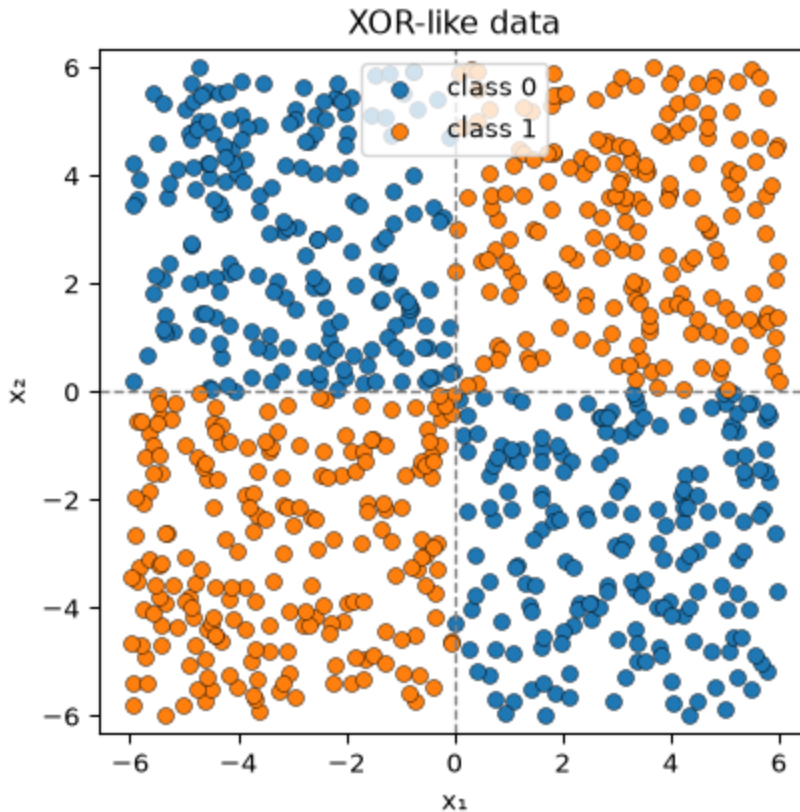
y = ((x1 * x2) > 0).astype(int)

plt.figure(figsize=(4.5, 4.5))
plt.scatter(X[y == 0, 0], X[y == 0, 1],
            color="C0", label="class 0", edgecolor="k", linewidth=0.3)
plt.scatter(X[y == 1, 0], X[y == 1, 1],
            color="C1", label="class 1", edgecolor="k", linewidth=0.3)

plt.axhline(0, color="gray", linestyle="--", linewidth=1)
plt.axvline(0, color="gray", linestyle="--", linewidth=1)

plt.xlabel("x1")
plt.ylabel("x2")
plt.title("XOR-like data")
plt.xlim(-6, 6)
```

```
plt.ylim(-6, 6)
plt.axis("equal")
plt.legend()
plt.tight_layout()
plt.show()
```



Would the same architecture, 2, 4, 4, 1, work for the XOR-like dataset?

XOR-like data (continued)

```
In [27]: X_train, X_test, y_train, y_test = train_test_split(
          X, y, test_size=0.25, random_state=42, stratify=y
        )

model = NaiveMLP([2, 4, 4, 1], step=0.06, seed=0)

print("Initial loss:", model.loss(X_train, y_train))

model.train(X_train, y_train, epochs=1000, verbose=False)

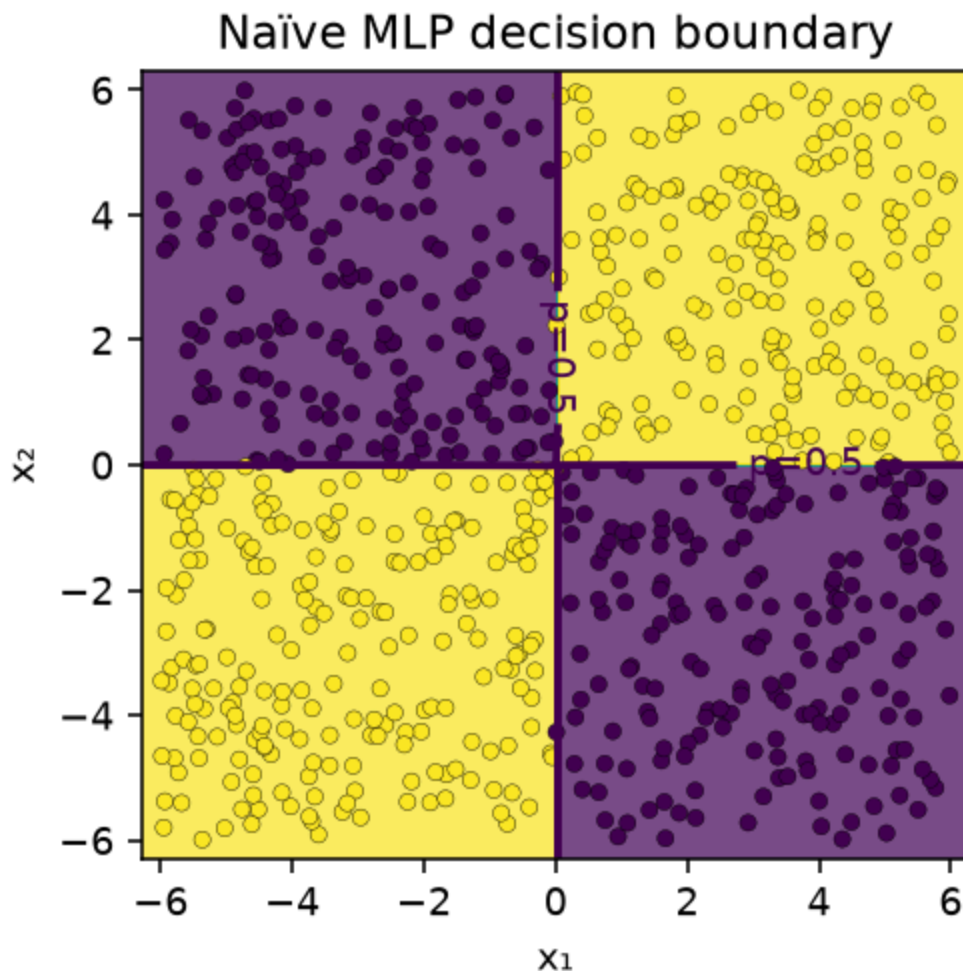
print("Final loss:", model.loss(X_train, y_train))

print("Train acc:", accuracy_score(y_train, model.predict(X_train)))
print("Test acc: ", accuracy_score(y_test, model.predict(X_test)))
```

```
Initial loss: 0.7029764953722529
Final loss: 3.4494531195516116e-07
Train acc: 1.0
Test acc: 0.99
```

XOR-like data (continued)

```
In [28]: plot_decision_boundary(model, X, y)
```



Our simple neural network, along with its naïve training algorithm, was evaluated on two distinct tests. Despite the simplicity of the model, it successfully learned significantly different decision boundaries without necessitating the engineering of additional features.

For the sake of simplicity and clarity in this example, we utilized the raw data without applying any scaling. However, in practical scenarios, it is customary to scale or normalize features prior to training neural networks or any models that rely on gradient-based optimization. Scaling facilitates faster and more stable convergence of training algorithms, although it introduces additional preprocessing and postprocessing steps. Since the primary objective here is to comprehend the training mechanism rather than optimize for efficiency, we have deliberately chosen to omit scaling in this instance.

Before training,

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

After training, when you want to visualize the decision boundary in the original feature space, you'd unscale the coordinates using `scaler.inverse_transform`.

Drawbacks

- Computational inefficiency.
- Scalability limitations.
- Fixed step size ($\pm\eta$) lacks adaptivity.
- Poor coordination of parameters.
- No directional or magnitude information.
- Lack of sophisticated optimizer features.
- Potential for over-fitting or poor generalisation.
- Computational inefficiency: trying every parameter with three deltas each epoch scales poorly as model size grows.
- Fixed step size ($\pm\eta$) lacks adaptivity: too small $\rightarrow$ very slow; too large $\rightarrow$ overshoot/oscillate.
- Poor coordination of parameters: each parameter updated ignoring interactions $\rightarrow$ slower convergence in coupled networks.
- Discrete local search (rather than derivative-based): no directional or magnitude information $\rightarrow$ many epochs needed, risk of zig-zagging or getting stuck.
- Scalability limitations: full loss evaluation per parameter change $\rightarrow$ infeasible for large datasets or many parameters.
- Lack of sophisticated optimizer features: no momentum, adaptive rates, regularization built in $\rightarrow$ weaker performance and reliability.
- Potential for over-fitting or poor generalisation: aggressive training on full-batch loss without built-in regularisation may tailor too much to training data.
- Limited insight into update magnitude: only $\pm\eta$ or 0 choices mean no fine-tuning of step size per parameter or epoch.

Notation

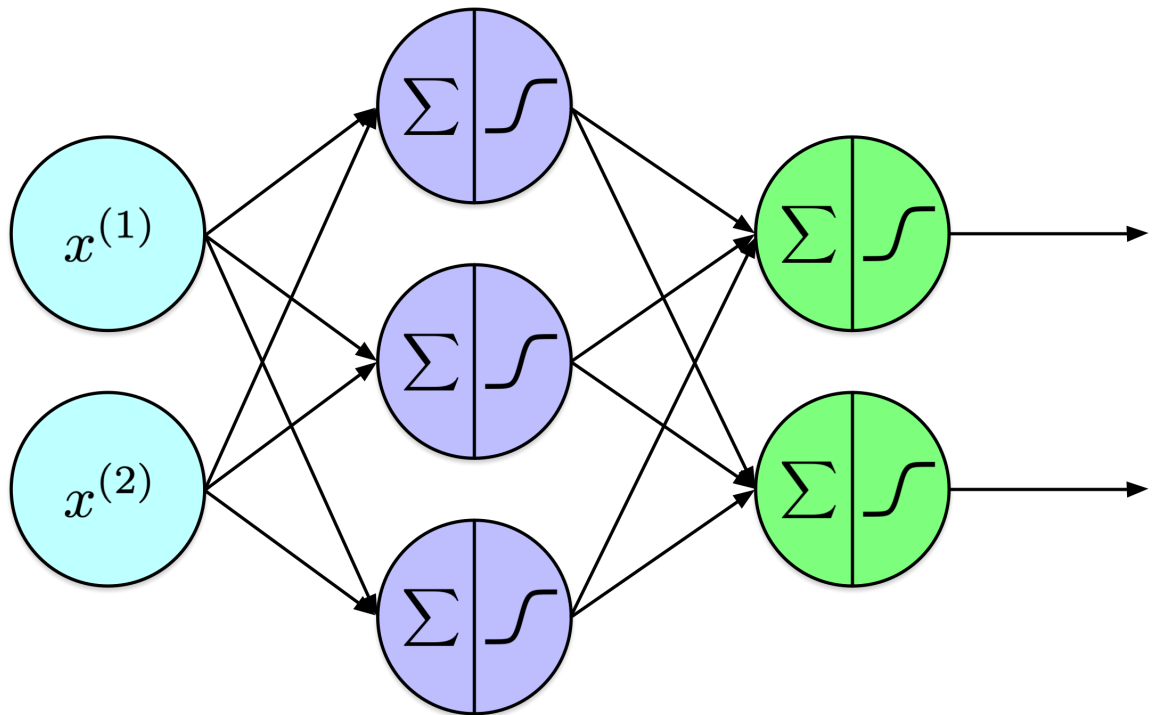
Notation

A **two-layer** perceptron computes:

$$\hat{y} = \phi_2(\phi_1(X))$$

where

$$\phi_l(Z) = \phi(W_l Z_l + b_l)$$



Where ϕ is an activation function, W a weight matrix, X an input matrix, and b a bias vector.

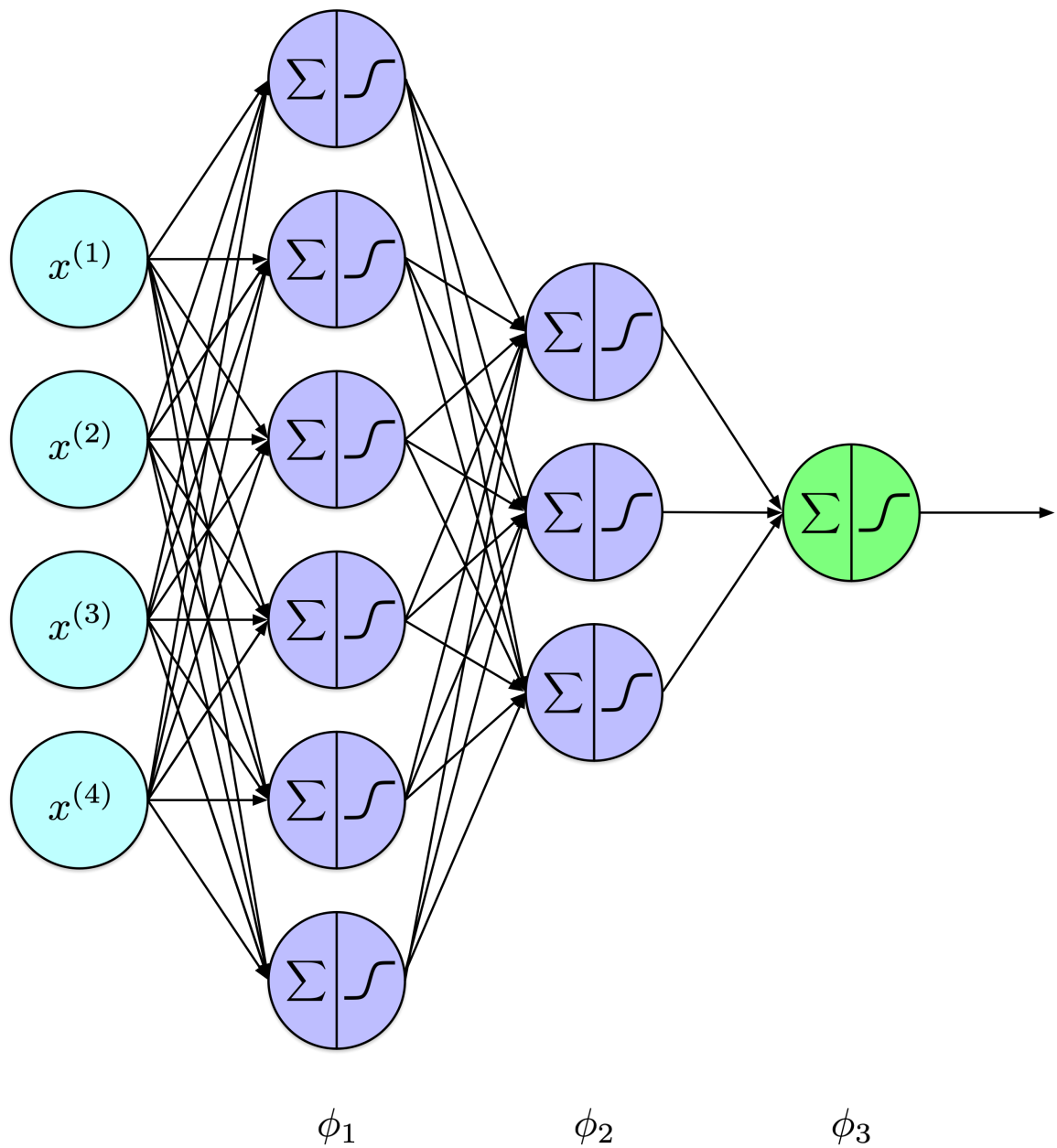
Notation

A **3-layer** perceptron computes:

$$\hat{y} = \phi_3(\phi_2(\phi_1(X)))$$

where

$$\phi_l(Z) = \phi(W_l Z_l + b_l)$$



Notation

A k -**layer** perceptron computes:

$$\hat{y} = \phi_k(\dots \phi_2(\phi_1(X)) \dots)$$

where

$$\phi_l(Z) = \phi(W_l Z_l + b_l)$$

A feed-forward network exhibits a consistent structure, where each layer executes the same type of computation on varying inputs. Specifically, the input to layer l is the output from layer $l - 1$.

In this notation, it is highlighted that only the first layer receives the input attributes directly. Each subsequent layer then processes the activations from the preceding layer, facilitating the learning of new representations. This underscores the consistent architecture of feedforward networks. As we will explore in the next lecture, increasing the number of layers does not inherently increase the algorithm's complexity.

Prologue

Summary

- Framed deep learning as layered function approximation across tasks.
- Described FNNs: inputs $\rightarrow$ hidden layers $\rightarrow$ outputs; information flowed forward only.
- Noted units used bias and activations; clarified why non-linearity mattered.
- Reviewed sigmoid/tanh/ReLU ranges and derivative behavior.
- Stated the Universal Approximation Theorem and its practical limits.
- Built a tiny MLP and computed predictions and BCE loss on toy data.
- Demonstrated a naïve, non-gradient training algorithm; it worked but scaled poorly and was brittle.
- Established compact layer notation, $\hat{y} = \phi_k(\dots \phi_2(\phi_1(X)) \dots)$ where $\phi_l(Z) = \phi(W_l Z_l + b_l)$, to prepare for backprop.

Next lecture

- We will introduce backprop, and discuss vanishing gradient, softmax, and regularization.

References

Cybenko, George V. 1989. "Approximation by Superpositions of a Sigmoidal Function." *Mathematics of Control, Signals and Systems* 2: 303–14.
<https://api.semanticscholar.org/CorpusID:3958369>.

Géron, Aurélien. 2022. *Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 3rd ed. O'Reilly Media, Inc.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. "Deep Residual Learning for Image Recognition." *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–78. <https://doi.org/10.1109/CVPR.2016.90>.

Hornik, Kurt, Maxwell Stinchcombe, and Halbert White. 1989. "Multilayer Feedforward Networks Are Universal Approximators." *Neural Networks* 2 (5): 359–66.
[https://doi.org/https://doi.org/10.1016/0893-6080\(89\)90020-8](https://doi.org/https://doi.org/10.1016/0893-6080(89)90020-8).

Russell, Stuart, and Peter Norvig. 2020. *Artificial Intelligence: A Modern Approach*. 4th ed. Pearson. <http://aima.cs.berkeley.edu/>.

Marcel **Turcotte**

Marcel.Turcotte@uOttawa.ca

School of Electrical Engineering and **Computer Science** (EECS)

University of Ottawa